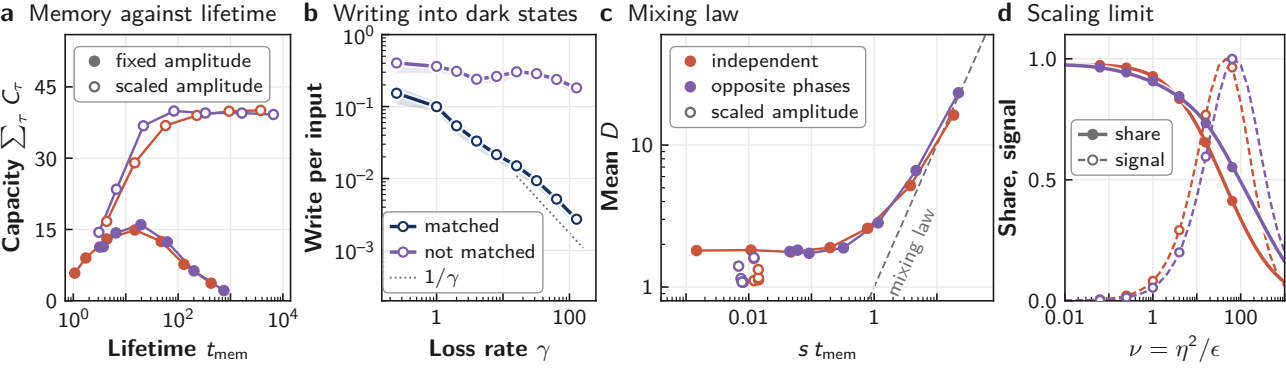

a Memory against lifetime
fixed amplitude
scaled amplitude
Capacity $\sum_\tau C_\tau$
45
30
15
0
10^0
10^2
10^4
Lifetime t_{mem}
b Writing into dark states
Write per input
10^0
10^{-1}
10^{-2}
10^{-3}
matched
not matched
$1/\gamma$
1
100
Loss rate γ
c Mixing law
independent
opposite phases
scaled amplitude
Mean D
10
1
mixing law
0.01
1
$s\,t_{\text{mem}}$
d Scaling limit
Share, signal
1.0
0.5
0.0
share
signal
0.01
1
100
$\nu = \eta^2/\epsilon$